LANDTECH INNOVATION HACKATHON 2026 · OFFICIAL SUBMISSION

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A Tamper-Evident Trust Layer Linking Deed Registration to Mutation in Bangladesh

Track: Digital Land Records, Verification & Digitization · Organized by Dept of CSE & CSE-TECH, Supported by Dept of Law, DIU

Chosen Track & Problem Statement	**P2 — Deed & Ownership Verification (Main Primary)**. Seamlessly integrated with **P3** (Fraud & Double-Selling Detection), **P4** (Smart e-Mutation & Registry Workflow), and **P5** (Citizen Pre-Purchase Verification) as a unified, full-lifecycle trust architecture.
Idea in One Line	Every registered deed generates a digitally signed block in an immutable per-plot hash chain; the plot is atomically locked until the AC Land office approves mutation; citizens and institutions can verify any deed copy in seconds without blind reliance on intermediaries.
Prototype Maturity	**Working prototype evaluated on 100% synthetic data:** 65 automated backend tests + Playwright E2E browser tests passing, independent offline Python verifier script, interactive attack lab, and localized AI Assistant. Unbuilt extensions are transparently flagged as *Planned*.

1. The Problem in Bangladesh: How Big, and Where It Breaks

Empirical Dimension	Verified Figures and Official Citations
Judicial Case Backlog	**4,516,603 pending cases** at 31 Dec 2024 (Supreme Court data) [1]. Scholarly analyses and Ministry of Land statements confirm land-related disputes represent **60% to 80%** of all civil litigations; the Land Secretary has publicly cited figures approaching 80% [1][2].
Households Impacted	**1 in 7 households** was directly embroiled in a land dispute in an empirical 14-district field survey (1,050 households); resolved cases required an average of **3 years**, while ongoing disputes averaged **8+ years** of litigation [3].
Corruption & Rent-Seeking	TIB National Household Survey 2023: **51% of households** seeking land services encountered corruption; **77.4% of citizens** seeking land mutation paid bribes. Average bribe: **Tk 12,719 for mutation** and **Tk 19,037 for deed registration**; Upazila Land Offices ranked worst in bribery incidence [4]. TIB 2025: Average land-service bribe per household stands at **Tk 11,311** [5].
National Operational Scale	**497 Sub-Registry Offices** registered **36.72 lakh deeds** in FY2017–18 [6]; over **22 lakh mutation petitions** are processed annually via the Ministry of Land e-Namjari gateway [7]. This represents the national volume a production software architecture must support.

Systemic Root Cause: A single land transaction in Bangladesh touches three institutional silos spanning two separate ministries: (1) the *Registration Directorate* (Ministry of Law) registers deeds and maintains manual volume books (*Balam*) under the *Registration Act 1908*; (2) the *AC Land Office* (Ministry of Land) updates the Record of Rights (*Khatian/Porcha*) via mutation under the *State Acquisition and Tenancy Act 1950*; and (3) the *DGLRS* conducts cadastral surveys across four historic generations (CS, SA, RS, BS). Land fraud thrives in the legal and administrative lag between these unlinked authorities.

Failure Point	Administrative Mechanism of Failure	Socio-Legal Impact
Forged Deed (Bia Dalil)	Fabricated chain deeds that never existed in the official *Balam* volume book look authentic on stamp paper. Altered copies circulate undetected.	Buyers, banks, and courts cannot verify authenticity on the spot; fraud is discovered years later in protracted lawsuits.
Double-Selling in the Lag	A plot is resold to a second buyer in the 30–60 day gap between sub-registry registration and AC Land mutation approval, while the seller is still on the record.	Multiple buyers claim the same parcel; creates immediate criminal trespassing and civil title suits.
Predatory Mutation Delays	Citizens must navigate separate offices and 3–4 manual visits; processing takes 28–60+ days, creating fertile ground for broker exploitation.	Buyers remain exposed without formal title; delays are opaque with zero automated accountability.
Archaic Citizen Asymmetry	Four overlapping survey systems (CS, SA, RS, BS), 16-anna fractional shares (*ana-gonda-kara-kranti*), and cursive Bangla script exclude ordinary citizens.	Rural farmers, seniors, and expatriates are forced into total dependence on unregulated brokers (*dalals*).

2. What Already Exists, and the Gap Bhumi Lipi Fills

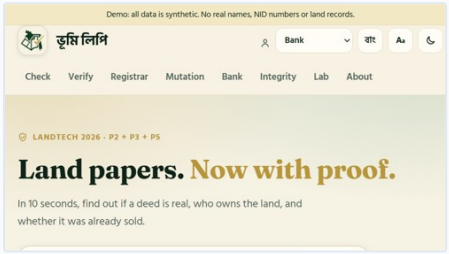
Bangladesh has made commendable progress through e-Mutation (*e-Namjari*), e-Porcha, the Digital Record Room, the Land Management Automation Project (LMAP/DLMS), and the ongoing Bangladesh Digital Survey (BDS) [9][10][11]. Recent 2026 directives introduced online seller phone/NID confirmation at registration [12]. **However, four critical architectural gaps remain unaddressed:**

> **The 4 Unsolved Gaps:** (1) *Independent Verifiability:* Outsiders cannot verify that a printed deed copy matches the registry without querying internal servers; (2) *Atomic Cross-Agency Lock:* No real-time lock blocks a second sale while mutation is pending; (3) *Tamper Evidence:* Centralized databases lack cryptographic guarantees against retroactive insider doctoring; (4) *Zero-Downtime Transition:* Existing offices cannot tolerate high-friction workflow replacements.

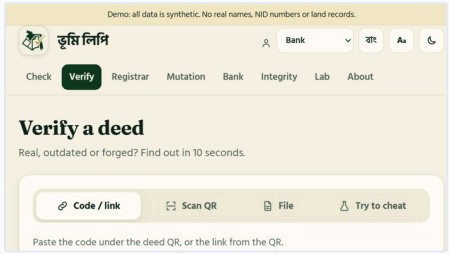
Bhumi Lipi fills this gap as a non-disruptive trust sidecar: It ingests a read-only stream from existing sub-registry and e-Mutation systems, binds records into an immutable per-plot chain, and provides instant public verification without forcing any office to abandon its active software.

3. The Proposed Solution: A Linked Cryptographic Lifecycle

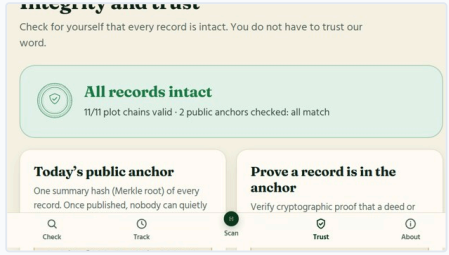
Lifecycle Step	Technical & Operational Process	Prevented Vulnerability
1. Register & Validate	Sub-registrar inputs deed details. **11 explainable heuristic rules** execute instantly (seller mismatch, khas/government land, court injunctions, rapid successive transfer, area mismatch).	Fraudulent sellers, disputed plots, and illegal transfers are blocked prior to formal deed execution.
2. Seal & Hash	Record fields are canonicalized and hashed (**SHA-256**), cryptographically linked to the plot's prior block, and signed with the office's **Ed25519 asymmetric key**. A tamper-proof QR seal is printed onto the physical deed.	Eliminates fabricated historical deeds and silent retrospective record alterations.
3. Lock & Auto-Mutate	Plot is atomically locked against secondary transactions. An e-Mutation case is spawned automatically with **8 automated validation checks** and a unique tracking ID. An SLA aging chip monitors processing days.	Completely stops double-selling during the administrative window; makes bureaucratic latency publicly visible.
4. Adjudicate & Settle	AC Land officer reviews from a prioritized queue. Upon approval, the plot unlocks and the new owner is committed as the new chain head. Rejections commit a signed `REVERSAL` block without deleting audit history.	Eliminates inter-agency record misalignment; ensures full auditable trace for contested claims.
5. Citizen Verify	Anyone scans the QR or uploads a digital deed file to receive an instant tripartite verdict: **Genuine**, **Outdated** (authentic prior deed, but superseded), or **Forged**. Simple mode provides voice read-aloud for elders.	Breaks citizen dependence on middlemen; empowers buyers and lending banks with instant, self-service verification.
6. Public Proof Anchor	Daily records are compiled into a binary **Merkle tree** whose root is published to an external public anchor. Citizens verify inclusion proofs in-browser or via an offline, independent Python CLI script.	Eliminates blind trust in hosting providers; mathematical proof replaces institutional assertions.



Plot Chain: Immutable sequential history of deed transfers



Verify Engine: QR scan, file-drop, and character-level diffs



Public Anchor: Merkle inclusion proofs & independent verifier

4. How We Sharpened the Idea After Studying the Registry

Field analysis of Bangladesh's land administrative jurisprudence revealed real-world edge cases where naive technical software fails. Each weakness has been addressed through defensive architectural hardening:

Identified Weak Point	Defensive Architectural & Institutional Remedy	Implementation Status
A Deed is Not a Title Under Bangladesh law, registration certifies execution, not title. A validly registered deed from a non-owner conveys zero ownership.	Every verification report strictly separates **Deed Integrity** (Genuine/Outdated/Forged) from **Record Agreement** (Does the deed's seller match the latest mutated Khatian and survey record?). Green indicates "*Records agree and no blocks found*", never "*Government guarantees title*".	**Built** (dual wording & mismatch flags); full split report *Planned*
Garbage In: Impersonation & Fake POA A digital signature cannot prevent an impostor with a forged Power of Attorney (*Am-Moktarnama*) from executing a sale.	Hash verified NID/biometric tokens at the registry. Mandate dual-officer maker-checker review for POA transactions. Dispatch automated SMS alerts to recorded landowners upon transaction initiation with an objection window.	**Built** (POA heuristic rule); owner alert & dual-signing *Planned*
Malicious Insider with Signing Key A compromised administrative official possessing a valid key could re-sign historical chains.	Daily public Merkle roots make retrospective chain alterations mathematically visible. Production specifies hardware security modules (FIPS YubiKeys), key revocation protocols, and threshold co-signing across 3 independent bodies.	**Built** (daily anchor & Attack Lab); hardware keys *Planned*
Mutation Latency & Rent-Seeking Digitization alone does not eliminate bureaucratic stalling or demands for speed money.	Enforce visible, political accountability: Real-time waiting-days latency chip; automated escalation hierarchy (UNO → ADC Revenue → DC); and a public Upazila efficiency leaderboard. Software escalates; it never auto-approves.	**Built** (aging chip); administrative escalation ladder *Planned*
Unverifiable Legacy Archives Centuries of British and Pakistani-era volume books cannot be retrospectively proven by software.	Avoid speculative back-verification: Establish a dated, officer-certified baseline seal of scanned *Balam* pages and Khatians. Any post-baseline modification becomes instantly detectable. Bangla OCR paired with human review.	*Planned (Phase B)*; synthetic baseline demo active
Parallel System Resistance Field offices will reject duplicate data entry across competing platforms.	Zero duplicate data entry: Day-1 deployment functions as a passive read-only adapter ingesting JSON feeds from e-Namjari and deed registration software, transitioning to bi-directional APIs.	**Built** (adapter schema); live departmental webhooks *Planned*
Marginalized Citizens & Data Privacy Rural citizens lack smartphones; publishing private NIDs compromises citizen data.	Public checks display only information already accessible on public Khatians (Plot, Mouza, Holder Name, Share); NIDs and phone numbers are salted and masked. Offline QR verification supported at Union Digital Centers (UDCs).	**Built** (field masking & offline QR); SMS/USSD fallback *Planned*

5. System Architecture, Tech Stack & Cryptographic Design

System Layer	Technology Selection, Architectural Role & Security Guarantees
Presentation Layer	**Vanilla JavaScript (ES6+), zero build tools:** Eliminates build vulnerabilities. Custom responsive CSS design system (dark/light mode, mobile bottom navigation, desktop drawers). Strict `textContent` bindings prevent Cross-Site Scripting (XSS). High-contrast Simple Mode for seniors with Bangla text-to-speech audio guidance and *ana-gonda* fractional share calculator.
Gateway & HTTP Layer	**Python Standard Library HTTP Engine:** Framework-free implementation minimizes external attack surface. Enforces token-bucket per-client rate limiting, 64 KB request payload caps, strict Content-Security-Policy (CSP), and path-traversal-safe static file routing.
Core Service Domain	**Clean Domain Architecture:** Decoupled modules for deed registration, plot verification, mutation lifecycle state machine, bank lien management, integrity anchoring, and localized AI assistance. Modular design allows replacing the gateway with FastAPI/PostgreSQL without altering business logic.
Cryptographic Engine	**Standardized Cryptographic Primitives:** SHA-256 per-plot hash chaining, asymmetric Edwards-curve digital signatures (**Ed25519** via `cryptography`), binary Merkle trees with $O(\log N)$ inclusion proofs, and an independent verification script executing without server dependencies.
Persistence & Immutability	**SQLite3 with Database-Level Triggers:** Immutability is enforced at the database layer; SQL triggers abort `UPDATE` and `DELETE` statements on sealed blocks, mutation logs, and anchors. Includes chain-guard triggers preventing plot forks. (Enterprise PostgreSQL migration ready).
Bhumi Lipi AI Assistant	**Grounded Legal Intelligence:** Localized retrieval assistant grounded in statutory Bangladesh land enactments (*State Acquisition and Tenancy Act 1950*, *Registration Act 1908*, *Transfer of Property Act 1882*). Answers citizen land questions in plain Bangla and English without hallucinations.

National Scale Mathematical Verification: Bangladesh registers ~36.72 lakh deeds annually [6] (~10,000 deeds/day). Across an 8-hour working day, this averages **1 deed every 2.9 seconds** nationally. SHA-256 hashing and Ed25519 signing execute in **<0.5 milliseconds** on commodity hardware. Bhumi Lipi effortlessly handles national throughput on a standard cloud instance or existing government server.

6. How Much of the Problem Does It Solve? (Impact Matrix)

No responsible software system can claim to eliminate 100% of land disputes, as many stem from physical possession or inheritance disputes. What can be precisely stated is which specific failure modes the cryptographic design reaches, and how a pilot would measure success:

Root Cause of Land Dispute	System Reach	Technical and Operational Elimination Mechanism
Forged or Altered Deeds	**Directly**	Every deed sealed post-adoption is verified in seconds against the Ed25519-signed hash chain. Forged or altered paper copies fail verification instantly.
Double-Selling of Land	**Directly**	Plot-locking during pending mutations, shared hash chains, and bank lien rules programmatically block secondary registrations on active parcels.
Citizen Information Asymmetry	**Directly**	Public self-service verification via QR, Simple Mode for seniors, plain-language legal assistant, and downloadable title search certificates eliminate middleman dependence.
Registration-Mutation Mismatch	**Mostly**	Unified lifecycle automatically triggers e-Mutation upon deed registration, eliminating record divergences between Sub-Registry and AC Land offices.
Discretionary Mutation Delays	**Partly**	Visible waiting-day latency chips, fewer administrative handoffs, and hierarchical escalation expose delays; institutional administrative enforcement remains necessary.
Survey Lineage & Area Mismatches	**Partly**	Automated area mismatch flags warn buyers; full resolution of historical survey discrepancies (CS vs RS vs BS) requires formal land survey tribunals.
Inheritance Disputes (Warish)	**Planned**	Succession share ledger is architected for Phase B; contested family shares require judicial succession certificates.
Illegal Possession & Encroachment	**No**	Physical boundary and law enforcement issues; software provides definitive cryptographic title evidence for courts but cannot police land physically.

Pilot Key Performance Indicators (KPIs): (1) Double-sale attempts blocked (target: 100%); (2) Forged deed detection rate (target: 100%); (3) Median days from registration to mutation completion (target: reduction from 28+ days to <10 days); (4) Citizen self-service verification checks per month.

7. International Benchmarking and Cost Feasibility

Bhumi Lipi synthesizes proven international models: Georgia's public ledger deed timestamping (NAPR), Estonia's KSI blockchain hash-calendar synchronization, India's Unique Land Parcel Identification Number (ULPIN), and Rwanda's nationwide land registration. **Unlike bloated public blockchain implementations, Bhumi Lipi incurs zero cryptocurrency transaction or gas fees:** Internal hash chains provide instant cryptographic proof, and only periodic Merkle roots are anchored. The entire software stack is open-source (MIT). Direct costs are confined to cloud infrastructure, hardware security keys (FIPS YubiKeys), and scanner equipment for legacy archival digitizations.

8. Plan for Countrywide Implementation (Phased Roadmap)

Stage	Administrative Scope	Key Deliverables, Institutional Tasks & Formal Exit Criteria
Stage 1: Pilot *(Months 1–6)*	1–2 Sub-Registries with matched AC Land Offices (e.g., Gazipur / Savar)	Deploy passive read-only event listener on existing deed registration software; issue hardware cryptographic keys (FIPS YubiKeys) to sub-registrars; staff training; baseline digital seal of one archival record room. **Exit Criteria:** Zero chain corruptions, 100% of double-sale attempts trapped, formal legal review completed.
Stage 2: District *(Months 7–15)*	All Sub-Registries and AC Land Offices in 1 District	Migrate to enterprise PostgreSQL cluster; role-based multi-factor authentication; automated citizen SMS alert gateway; public Upazila latency delay leaderboard; offline QR verification at Union Digital Centers (UDCs). **Exit Criteria:** Mutation processing delays measurably reduced; zero undetected deed forgeries.
Stage 3: Division *(Months 16–27)*	Multiple Districts across 1 Administrative Division	Bi-directional API bridge to Ministry of Land e-Namjari; national identity integration (Election Commission NID verification); three-party threshold anchor co-signing between DC Office, District Judge Court, and Public University. **Exit Criteria:** Formal inter-ministerial data-sharing agreements active; inter-district fraud attempts blocked.
Stage 4: National *(Months 28–42)*	All 497 Sub-Registries and 500+ AC Land Offices in Bangladesh	Nationwide shared public Merkle anchor; public open-source verification portal; scheduled retrospective digitization of legacy Khatians and volume books using Bangla OCR with mandatory gazetted officer review. **Exit Criteria:** Universal countrywide deed verification; complete elimination of land double-selling across Bangladesh.

9. Data Privacy, Security Guardrails & AI Tool Disclosure

> **Data, Privacy & Security Compliance (Section 8 of Rule Book):** All project demonstrations utilize **100% synthetic (dummy) data**. No real personal names, national identity numbers (NID), phone numbers, physical addresses, or actual government land records are used, stored, or displayed. Absolutely no scraping, automated querying, or bulk downloading was conducted on any government portal. Zero penetration testing was attempted on live government systems; all security testing and attack simulations are conducted strictly within the team's local prototype Attack Lab.

Disclosure of AI Tools (Section 9 of Rule Book): In accordance with Section 9 of the LandTech 2026 Rule Book, the team fully discloses the use of advanced AI coding assistants during system conception, engineering, and testing:

- **Google Gemini (Gemini Antigravity):** Autonomous AI pair-programmer for cryptographic state modeling (SHA-256 per-plot hash chains, Ed25519 signing keyrings, Merkle proof calculation), responsive UI engineering (mobile bottom navigation, drawer modals, CSS design tokens, WCAG-compliant high-contrast themes), domain prompt engineering for the localized Bhumi Lipi AI Assistant, and automated regression test design.
- **Anthropic Claude (Claude 3.7 Sonnet / Claude Code):** System specification authoring (`docs/SPEC.md`), statutory legal analysis of Bangladesh land governance frameworks (comparing *State Acquisition and Tenancy Act 1950* with *Registration Act 1908*), threat modeling against MITRE ATT&CK style tamper scenarios (`docs/THREAT_MODEL.md`), and unit test edge-case generation.
- **Student Code Mastery Guarantee:** Both student team members independently reviewed, debugged, executed, and verified 100% of the code, database triggers, cryptographic routines, and architectural choices. Both student members are fully capable of explaining and defending every technical component before the judging panel.

10. Team and Mentor Details

Team Role	Member Name	DIU ID / Affiliation	Contact Details
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Team Name: Bhumi Lipi | **Project Repository:** [GitHub / GitLab Link] | **Live Prototype URL:** [Render Deployment Link]

References & Legal Sources: [1] Supreme Court of Bangladesh, Case Backlog Statistics (as of 31 Dec 2024); Dhaka Tribune, "Whose land is it anyway?". [2] Daily Star, Ministry of Land administrative updates and scholarly estimates (60%–80%). [3] PRI & BRAC, "One in 7 households facing land disputes", Empirical 14-district study (2014). [4] Transparency International Bangladesh (TIB), National Household Survey: Corruption in Service Sectors 2023, via The Financial Express. [5] TIB, National Household Survey 2025 Executive Summary (Average bribe per household: Tk 11,311). [6] Daily Star on Directorate of Registration & TIB Sub-Registrar report (FY2017–18). [7] Ministry of Land, e-Namjari Annual Application Statistics (over 22 lakh applications/year). [8] Practitioner guides on Bangladesh Land Law (Transfer of Property Act 1882, Registration Act 1908, SATA 1950). [9] UN DESA, Bangladesh e-Mutation Public Innovation Study (2020). [10] Ministry of Land, Land Management Automation Project (LMAP/DLMS). [11] The Business Standard, Bangladesh Digital Survey (BDS) Cadastral Report. [12] Gonobarta, Official press briefing by Land Secretary on 2026 registration reforms.